VICTORY 2027

PHOTOSYNTHESIS
(PHOTOSYNTHESIS PROCESS)

LECTURE- 03 | BIOLOGY

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TOPICS TO BE COVERED

1. Factors affecting Photosynthesis
2. Experiments on Photosynthesis

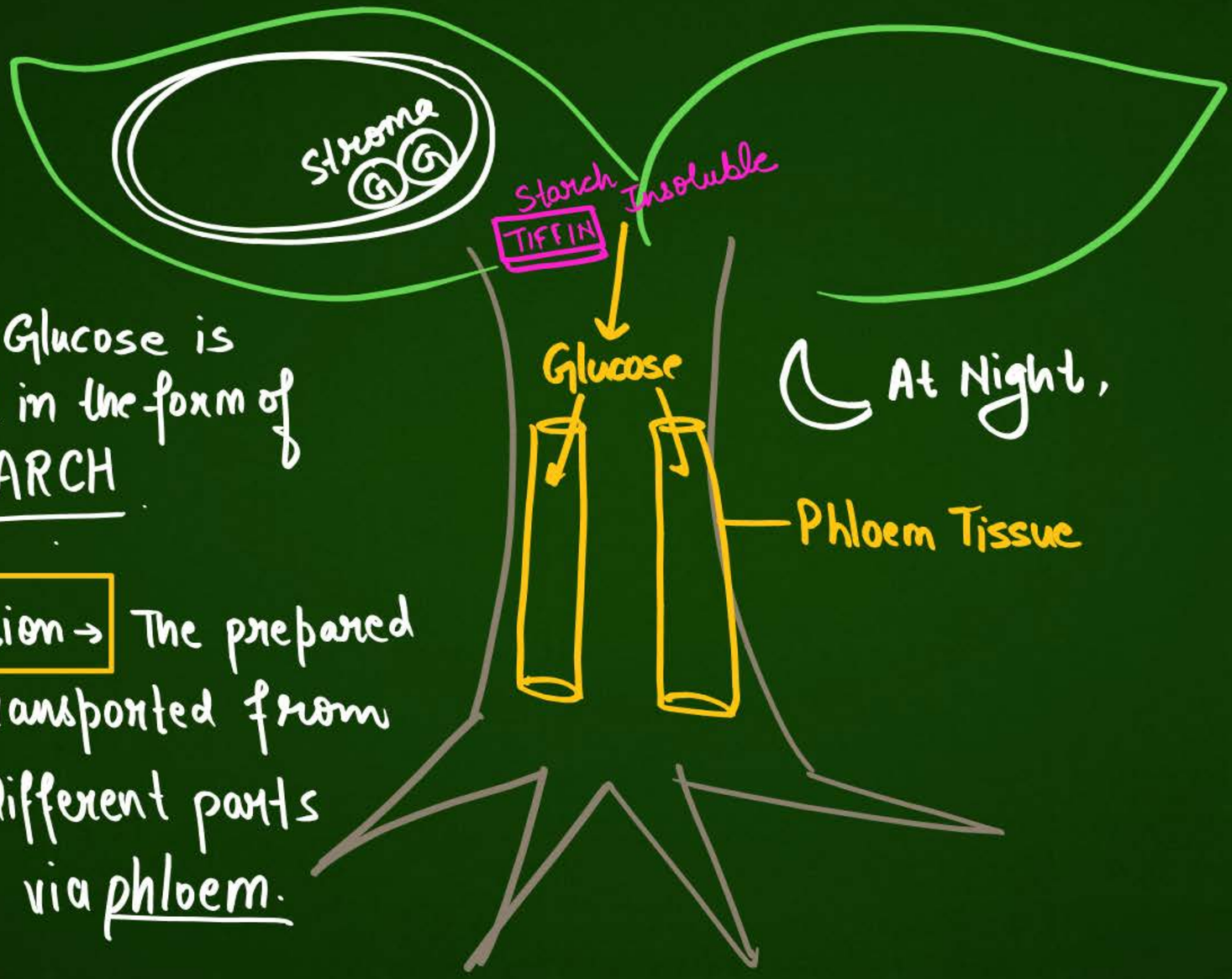

HELLO
NINJAS
忍者参上
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忍者参上

Light-independent phase (Biosynthetic Phase)

- The dark reactions use the ATP and NADPH produced during the light reactions for energy
- Does not depend on light
- Occur in stroma of chloroplast
- Carbon dioxide is converted into Glucose.
- Also known as Calvin cycle

$$6CO_2 \xrightarrow[\text{Cycle}]{\text{Calvin}} C_6H_{12}O_6$$

Glucose



Extra Glucose is stored in the form of STARCH

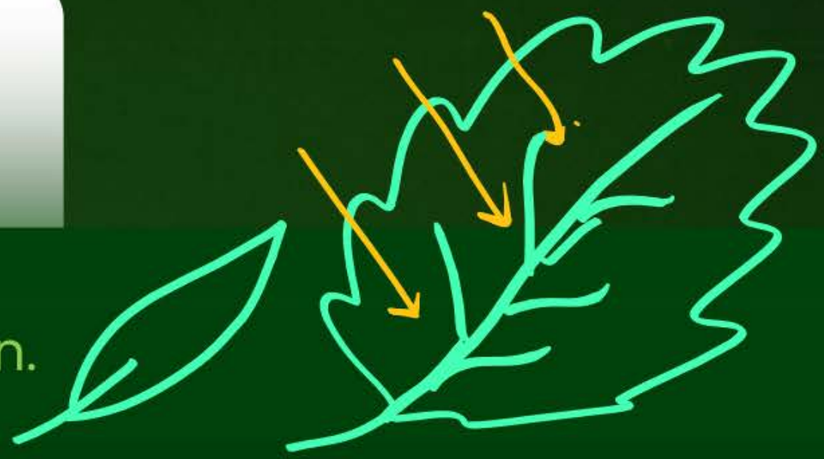
Translocation → The prepared food is transported from leaf to different parts of plant via phloem.

Utilization of Synthesized Food &its Translocation :

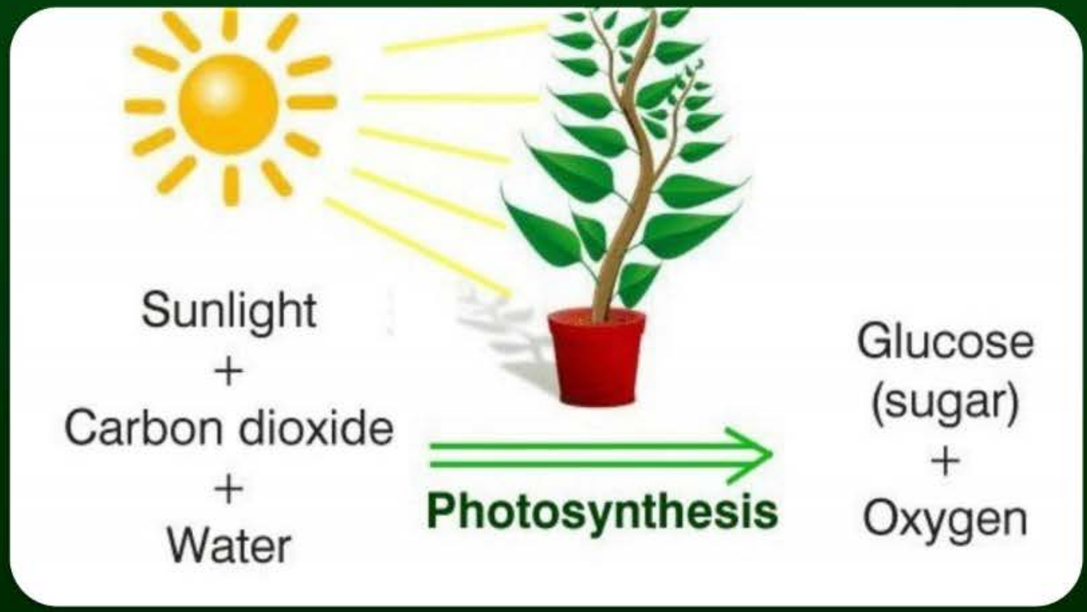
- **Polymerisation of Glucose** G-G-G-G-G-G-G-G- ....

- The glucose is formed in the leaf very rapidly during photosynthesis and it cannot be transported to other parts with the same rapidity.
- So, it is converted into insoluble starch for temporary storage in the leaf.
- At night, the starch is reconverted into soluble sugar which is transported in solution through the veins of the leaf and down through the phloem of the stem.
- Some plants also form SUCROSE(cane sugar) or some into oils

Adaptations in Leaf to Perform Photosynthesis



- Large surface area for maximum absorption.
- Leaf arrangement is at the right angle.
- Cuticle and epidermis are thin.
- Numerous stomata for exchange.
- Thin leaves for rapid transport.
- More chloroplasts on upper surface.
- Extensive vein system.



BIOLOGY

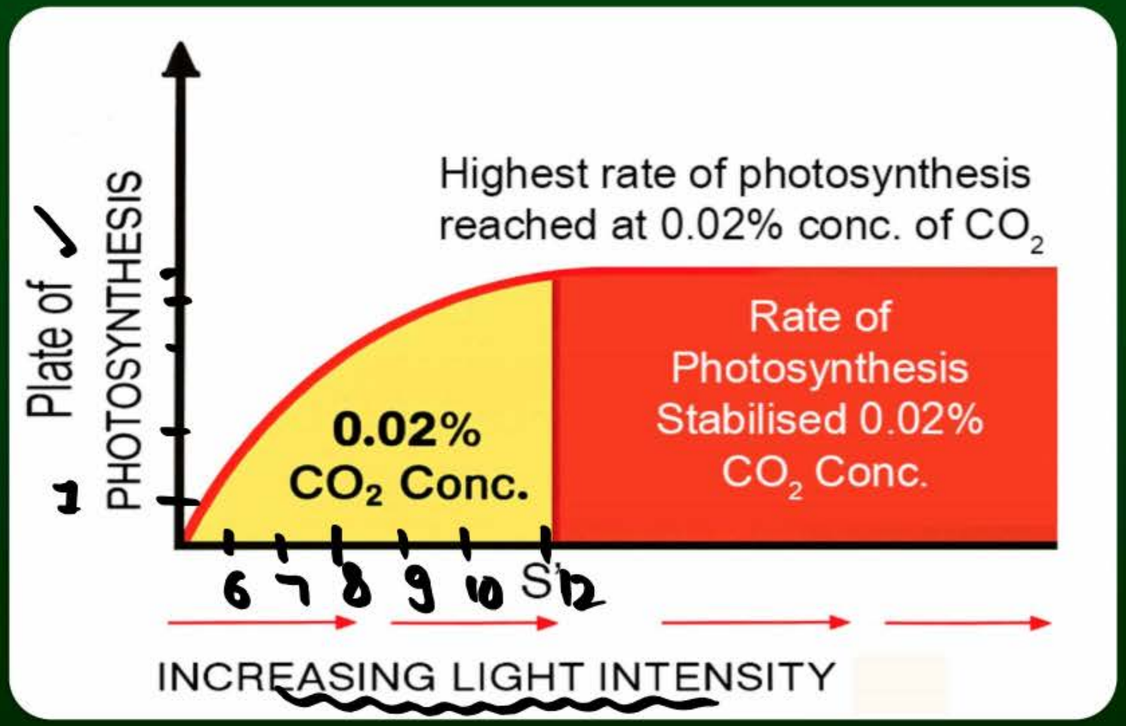
Factors Affecting Photosynthesis

❑ EXTERNAL FACTORS

(i) Light intensity

(ii) Carbon dioxide concentration

- Photosynthesis increases with increase in light intensity but upto a certain limit, It gets stabilized at 0.02% CO_2.



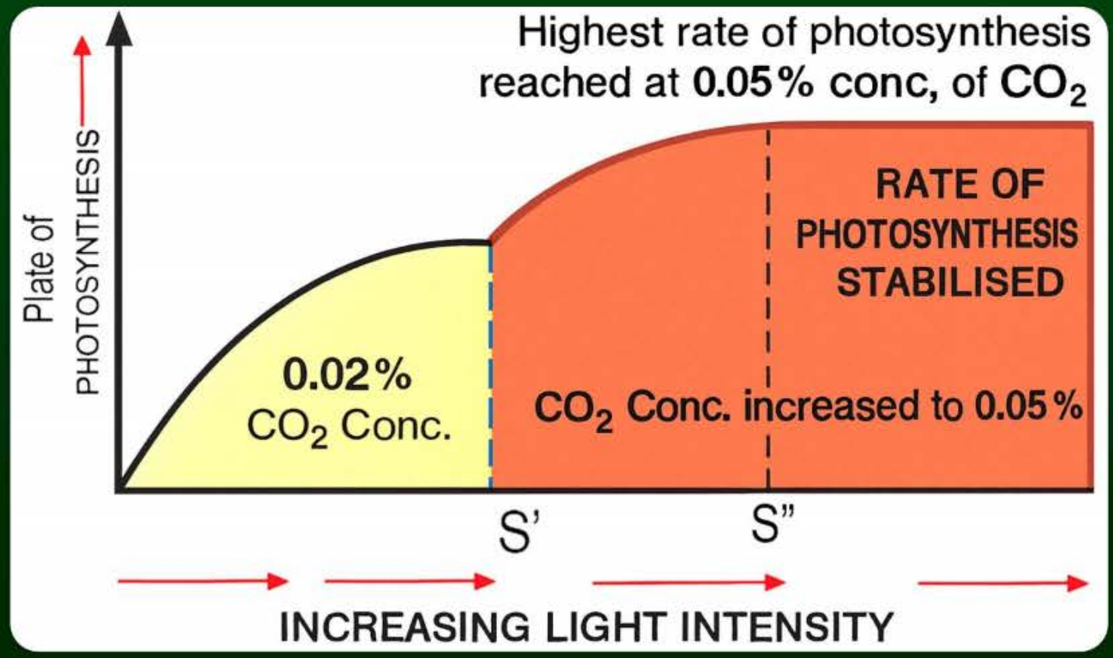
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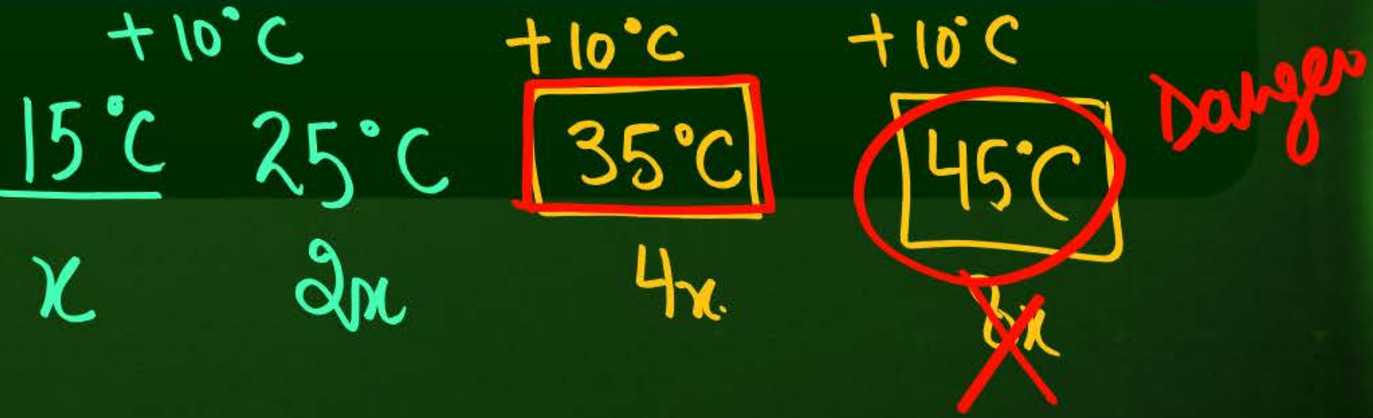
- But if CO_2 increases at 0.05% photosynthesis increases with increase in light intensity.



Factors Affecting Photosynthesis

(iii) Temperature

- The rise occurs with optimum temperature and max suitable temperature - 35 degree Celsius.
- A rise of 10 degree C up to the optimum temperature doubles the rate of photosynthesis.



(iv) Water content

The less water content will reduce the rate of photosynthesis

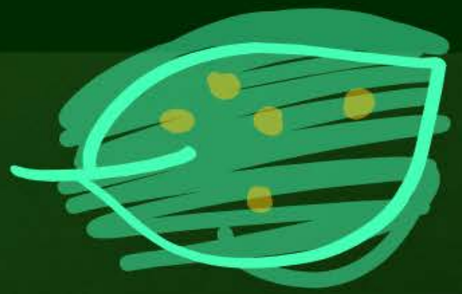
INTERNAL FACTORS

(i) **Chlorophyll -** Nutritional deficiencies of minerals cause loss of chlorophyll, and hence the drop in trapping solar energy.

C, H, O, N, Mg.

(ii) **Protoplasm -** Dehydration of protoplasm for some reason reduces the rate of photosynthesis. Similarly, the accumulation of carbohydrates (sugar and starch) also reduces the rate of photosynthesis.

(iii) **Structure of leaf -** The thickness of cuticle, the distribution of stomata and the size of the leaf influence the amount of light and the amount of CO2 entering the leaf.



QUESTION

Are the following statements true or false? Give reason in support of your answer.

(a) The rate of photosynthesis continues to rise as long as the intensity of light rises. but upto a certain limit.

A) True ✓

B) False



QUESTION

Are the following statements true or false? Give reason in support of your answer.

(b) The outside atmospheric temperature has no effect on the rate of photosynthesis.

A) TRUE

B) FALSE ✓



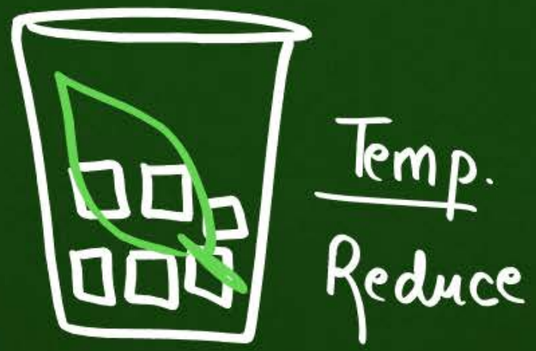
QUESTION

Are the following statements true or false? Give reason in support of your answer.

(c) If you immerse a leaf intact on the plant in ice cold water, it will continue to photosynthesise in bright sunshine.

A) True

B) False





Photosynthesis at Night

(i) Some desert plants and epiphytes are able to release oxygen even during the night as they perform photosynthesis using a different pathway called the CAM pathway Crassulacean acid Metabolism.

Examples of such plants include Aloe vera, Tulsi, Neem, Peepal, etc.





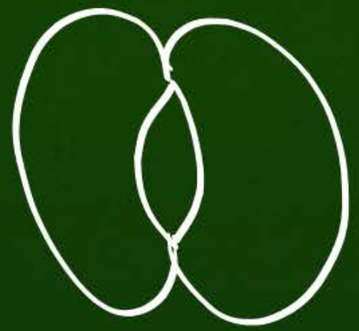
Guard Cells are present around ____

A) Root hair

B) Lenticels

C) Thylakoid

D) Stoma



'Destarching of leaf.' → (when the leaf has used up all the prepared Starch)

24-48 hours in dark room.

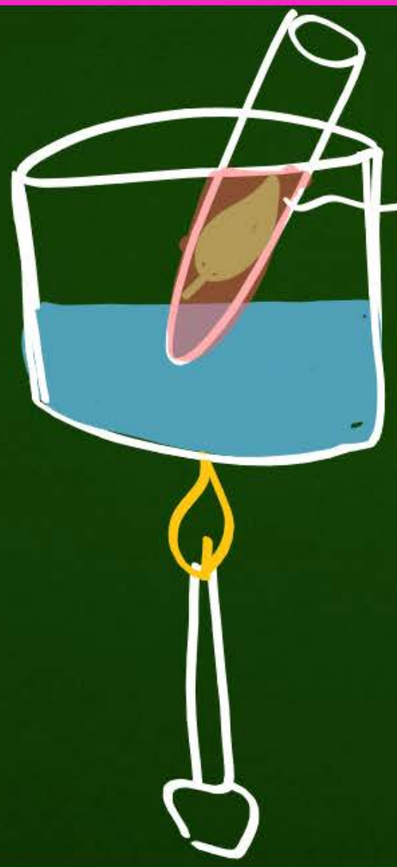
Iodine

↳ ★ leaf [starch]

↓

Blue-black colour

→ ★ Destarch leaf (Brown Colour)



Boiling in 'Methylated Spirit'

↓

To remove chlorophyll (The leaf turns colourless and brittle (hard)

↓

To soften it add it in hot H_2O

→ Add Iodine → No Blue Black

Thank You